

BIOLOGICAL SCIENCE Part 14

1. Which biological classification approach categorize organism based on derived characteristics traced to a group's common ancestor?

- A. Chemotaxonomy
- B. Cladistics Taxonomy
- C. Cystaxonomy
- D. Numerical Taxonomy

Ans: B. Cladistics Taxonomy

- Cladistics is a branch of biology that determines the evolutionary relationship between organism based on derived characteristics that can be traced to group's most recent common ancestor. Willi Hennig is considered as the founder of cladistics. Chemo taxonomy is a system classification based on biomchemical characteristics like amino acids, proteins and DNA sequence, Cytotaxonomy is classification of organisms based on cellular Structure and function most particularly in the number of chromosomes. Numerical Taxonomy is the quantitative evaluation of resembles and difference or primitiveness and advancement through statistical methods

2. Plants are classified as herbs, shrubs and trees. What is the basis in this biological classification?

- A. Evolutionary relationship
- B. Gene pool
- C. Morphology
- D. Phylogenetics

Ans: C. Morphology

-Morphology forms the primary basis for classifying organism into various taxonomic groups of taxa. It deals with the structure or the physical appearance of the organisms. The similarities in the morphological characters are used for grouping the plants together as herbs, shrubs, vines and trees. Phylogenetics is the study of evolutionary relationship among the species.

3. How is species defined?

- A. Based on the ability to interbreed
- B. Based on the external appearance
- C. Based on the DNA and protein sequences
- D. Based on the number of adaptation to the same

Ans: A. Based on the ability to interbreed

- The general accepted criterion for defining a species is that organisms of the same species interbreed under natural conditions to yield fertile offspring. Individuals

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of different species normally do not mate. Reproductive isolation defines a biological species.

4. Which of the following taxonomic ranks is properly arranged from most inclusive?

- A. Species, genus, family, order, class, phylum, kingdom, domain.
- B. Kingdom, phylum, domain, family, class, order, genus, species
- C. Domain, Kingdom, phylum, family, class, order, genus, species
- D. Domain, kingdom, phylum, class, order, family, genus, species

Ans: D. Domain, kingdom, phylum, class, order, family, genus, species

-Each organism can be classified at eight taxonomic ranks: domain kingdom phylum class, order, family genus species. The ranks form a nested hierarchy each domain includes a number of kingdoms; each kingdom contains a number of phyla; each phylum contains a number of classes and so on. Each successive group contains fewer organism but the organism are more similar. An easy way to remember the hierarchy is to use the mnemonics: Did King Philip Come Over For Great Spaghetti?

5. What is the equivalent of a division in plant taxonomy?

- A. Family
- B. Genus
- C. Kingdom
- D. Phylum

Ans: D. Phylum

-Originally the term division was used in botany instead of phylum but in 1933. The international Code of Nomenclature for algae, Fungi, and plants accepted the use of Phylum.

6. Eukaryota consist of organism that we have a true nucleus and structure called organelle that are surrounded by membranes. Which of the following are included in the Domain Eukarya?

- I. Animals and Plants
- II. Microbes
- III. Fungi and Protists
- IV. Bacteria

- A. I only
- B. I and III
- C. II and IV
- D. II, III and IV

Ans: B. I and III

- Eukaryota is one of the three domains of life containing organism whose cells have membrane-bounded organelles, especially the nucleus. Bacteria and Archaea are

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prokaryotes that have none of the characteristics. The kingdom most associated with Eukarya are Protista, Fungi, Plantae and animalia .

7. Who is considered to be the “first taxonomist”? He wrote De plantis, a work the contained 1500 species.

- A. Bauhin
- B. Caesalpino
- C. Dioscorides
- D. Theophrastus

Ans: B. Caesalpino

-Caesalpino(1519-1603) an Italian botanist who is sometimes called “the first master taxonomist” In 1583, he wrote De plantis, a work that contained 1500 species. His classification was based on growth habit together with the fruit and seed form rather than alphabetically or by medicinal properties.

Gaspard Bauhin and Caspar Bauhin are swiss brothers who wrote the work Pinax Theatri Botanici in 1623. The word “pinax” means register and the work is a listing of 6000 species. The bauhin brothers recognized genera and species as major taxonomic levels. Dioscorides (40-90 AD) was a Greek physician who wrote “De Materia Medica” a work consisting of about 600 plant species classified based on their medicinal properties. Theophrastus (370-285 BC) was a student of Aristotle and Plato. He wrote a classification of all known plants, De Historia Plantarum, which contained about 480 species . His classification was based on growth form. Linnaeus called him the “father of botany”.

8. What was the outstanding contribution of the two US zoologists namely, Kevin de Queiroz and Jacques Gauthier?

- A. They founded the Cladistic Era
- B. They initiated the zoological code
- C. They laid the foundation of the PhyloCode
- D. They started the construction of evolutionary trees.

Ans: C. They laid the foundation of the PhyloCode

-Two US zoologists by the name of Kevin de Queiroz and Jacques Gauthier initiated the discussions on a new nomenclature code for all organisms called the PhyloCode. The PhyloCode suggests a philosophical shift from naming species and subsequently classifying them to naming both species and subsequent classifying them naming both species and clades. Up to now the PhyloCode remains as a draft. It is controversial and highly debatable.

9. Who was the Swedish naturalist who introduced the two-part scientific name.

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- A. Carolus Linnaeus
- B. Charles Darwin
- C. Georges-Luise Leclerc de Buffon
- D. Oto Knutze

Ans: A. Carolus Linnaeus

-Carl von Linee (1707-1778) or Carolus Linnaeus was a Swedish naturalist who authored the global flora species plantarum published in 1573 and the tenth edition of Systema Naturae in 1758 including global fauna which introduced a binary form of species names called "trivial names" for both plants and animals. His works were regarded as the basis of modern taxonomy.

Charles Darwin linked species diversity to evolution. He hypothesized that classification would come to resemble genealogies (evolutionary theory)

Georges-Louis Leclerc de Buffon was a French naturalist who was a known critic of Linnaeus' works. He wrote "Histoire Naturelle", a 44 volume encyclopedia describing everything known about the natural world, wrestled with similarities of humans and apes. Originally, he intended to cover all three "kingdoms" of nature but the "Histoire Naturelle" ended up being limited to the animal and mineral kingdoms

Otto Kuntze was a German botanist who entirely revised the taxonomy in his work "Revisio Generum Plantarum" (1891) by changing 1000 generic names and 30,000 species names his work was widely rejected.

10. An organism is heterotrophic multicellular whose cells have cell wall. To which kingdom must it belong?

- A. Animalia
- B. Fungi
- C. Monera
- D. Plantae

Ans: B. Fungi

-Members of the kingdom Fungi are eukaryotes whose cells have cell walls. Mushrooms, molds, yeast and mildew are all examples of organisms under this kingdom. They are considered heterotrophs since they obtain nutrition by releasing digestive enzymes into a food source and absorbing the digested nutrients.

11. Under which kingdom are slime molds and algae classified?

- A. Animalia
- B. Fungi
- C. Monera
- D. Protista

Ans: D. Protista

-The kingdom Protista is composed of unicellular and multicellular eukaryotes it included all microscopic organisms that are not bacteria not animals not plants and not

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fungi, Slime molds, Water molds, water molds, diatoms dinoflagellates and algae are all protists.

12. Which of the following BEST describes an organism belonging to Kingdom Monera?

- A. it is unicellular eukaryote
- B. it is unicellular prokaryote
- C. It is multicellular heterotroph
- D. It is multicellular, autotrophic eukaryote

Ans: B. it is unicellular prokaryote

-The kingdom Monera includes all prokaryotes that are single-celled, most widely known as bacteria. These cells do not have nuclear membrane the chromosome is a single and circular, they also lack membrane bound cellular organelles. The kingdom is divided into two groups Archaeobacteria and Eubacteria

A protist is a unicellular eukaryote. An animal is a multicellular heterotroph. A plant is multicellular autotrophic eukaryotes.

13. How does a dichotomous key help in classifying living things?

- A. It presents the genus name and species name
- B. it provides the two best traits of the organism
- C. It gives the distinct structure of the organism together with its function.
- D. It shows a series of opposing description of basic features of an organism

Ans: D. It shows a series of opposing description of basic features of an organism

-A dichotomous key is tool used for identifying organism. It lists specific recognizable characteristics of many organisms. For each characteristic, the key list two contrasting options.

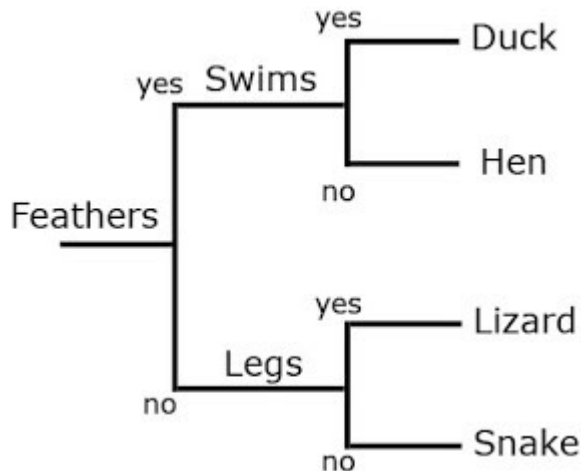
14. Using the dichotomous key below, how do you describe a duck?

- A. A feathered animal that swim
- B. A Feathered animal that cannot swim
- C. A non-feathered animal that can swim
- D. A non-feathered animal has no legs

Ans: A. A feathered animal that swim

- Using the dich key, a duck used for identifying organism. A hen is a feathered animal that cannot swim. A lizard is a non-feathered animal with legs. A snake is non-feathered animals without legs.

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15. Analyze the following animals: fish. Whale, house-fl, bat, penguin and mosquito. Using a dichotomous key, which will give you the best description?

- A. Bat,-fly, only at night
- B. House-fly only at night
- C. Mosquito- fly only at night

Ans: A. Bat,-fly, only at night

-Using the ditchotomous key, a bat can fly and does it at night. A house fly and mosquito can fly in any part of the day. Penguin is a flightless bird.

16. Which of the following is NOT always true about living things?

- A. Living things are single-celled organism
- B. Living things maintain internal consistency.
- C. Living things reproduce sexually or asexually
- D. Living things take in energy and use if for maintenance and growth

Ans: A. Living things are single-celled organism

- Choices B, C and D are always true, There are living things which are single-celled. They have everything they needed to be self-sufficient but many living things are multicellular. All animals, land plants and most fungi are multicellular. They consist of more than one cell.

17. Why do plants use different carbon fixation pathways?

- A. To minimize photorespiration
- B. To produce more glucose and oxygen
- C. The C_3 cycle is not very efficient in making food
- D. The C_4 cycle and CAM are very efficient in producing glucose

Ans: A. To minimize photorespiration

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- ON usual problem that many plant encounter is the washing of much solar energy known as photorespiration. To adapt to photorespiration there are alternate pathways for carbon fixation namely; C_4 and CAM (Crassulacean Acid Metabolism)

18. Which do animals need from their environment in order to survive?

- I. Air
- II. Food
- III. Light
- IV. Soil
- V. Water

Ans: C. I,II and IV

-animals needs air, food and water, light and soil are more needed by plants

19. Which plant is found along the seashore or brackish water in the Philippines whose milky juice poisonous and can cause blindness when it touches the eyes and blister to the skin?

- A. Buta-Buta
- B. Pongpong
- C. Talumunay
- D. Trompeta

Ans: A. Buta-Buta

- Buta-buta is a subtropical evergreen shrub with a woody stem whose milky juice sap can cause blindness to the eye and blister the yields a potent poison often used for suicide. Talumpanay is known for its hallucinogenic properties. It is often used to poison others by mixing it in tea. Trompeta or angels trumpet is a beautiful plant that can be deadly for it can cause accelerated heart rate which can be fatal.

20. The Talavera family lives in a Barrio, butch=, the youngest in the family is suffering from eczema. Which among the following may be used as medicinal plant to address his condition?

- A. Akapulko
- B. Niyog-niyogan
- C. Sambong
- D. Yerba Buena

Ans: A. Akapulko

- Akapulko is known as "ringworm" bush". It is used to treat ringworm, itchiness eczema and scabies. Niyog-Niyogan is used to eliminate intestinal parasites. Sambong is a medicinal plant and aches due to rheumatism and gout.

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